29/10/2

Lab 5 : Cuckoo Search (CS)

Cuckoo Search

Pseudocode :

Begin

Define the objective function f(x)

initialise parameters n, pa, max_iterations

generate n random nests

evaluate the fitness of each nest

best ← best fitness among all

iter ← 1

while iter <= max_iterations

for i = 1 to n do

generate random nest using levy flight function

new_fit ← evaluate fitness of new nest

if (new_fit > fitness) then

nest[i] = new nest

end_if

end_for

Re Arrange the nests on basis of the fitness

abandon the least fit nests upto pa factor

FOR each abandoned nest

generate new random nest using levy flight function

update the ones having better fitness

end for

Check for any new nests having better fitness

iter ← iter + 1

End while

Returns the best nest and its corresponding fitness

```
def cuckoo_search(n, pa, max_iter):
    n = initial n
    pa = cutoff factor
    max_iter = max iterations
    nests = arr(n)
    values = arr(n)
    for i = 1 to n do:
        x = random.randint()
        nests.append(fn(x))
        values.append(eval_fitness(nests[i]))
    iter = 1
    best = values.max()
    while iter <= max_iter
        for i = 1 to n do:
            x = nest[i].value()
            fnx = fn(x)
            v = eval_fitness(fnx)
            if (v > value[i])
                nest[i] = fnx
                best = max(best, v)
            end if
        endfor
        for j = n-1 to pa
            nests.remove(nests[j])
            y = nest[j].val
            fny = fn(y)
            vy = eval_fitness(fny)
            if (vy > best)
                best = max(best, vy)
                nests.append(fny)
            end if
        end for
```

POWER Generat

iter = iter + 1

end while

return nest[best] and value[best]

nest fitness

Paper: Modified Cukoo Search Algorithm to solve economic power dispatch optimization problems

Authors: Jian Zhao, Shimin Liu, Mengchu Zhou, Xiwang Guo

Summary:

The paper proposes a Modified Cukoo Search Algorithm (MCSA) to solve the economic dispatch problem in power systems. it main aim is to minimize the total fuel cost of power generation while satisfying electricity demand and generator operating constraints. This improves standard cukoo search. using adaptive step sizes, neighborhood search, and improved lamba search strategy

Objectives:

- Minimize total fuel cost of power generation
- improve cukoo search using modified optimization strategies
- Satisfy power demand and generator operating constraints
- Compare Proposed algorithm methods with other optimization methods.

Code :

. repair(P) : adjust generator output to satisfy power demand

• Cuckoo_search (seed, modified, nests, iteration) - optimize generator outputs using CSA or MCSA

• Cuckoo_search () : returns best power schedul, min cost, convergence history

• np.random.default_rng(seed) : Generates reproducible random numbers

• np.clip () - restricts generator o/p with ranges

• np.argsort () - identify nest with highest & lowest

o/p

Economic Power dispatch results

required demand : 850 MW

Generator outputs :

Generator	CSA (mw)	MCSA (MW)
1	374.42	374.75
2	189.89	189.63
3	285.69	285.62

Cost difference : 0.9350 units / hour

cost reduction : 0.0159 %.

CDR
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